

INCIDENT REPORT

SOC Home Lab — Simulated Threat Detection Exercise

Report ID: IR-2026-001 **Classification:** CONFIDENTIAL — INTERNAL USE ONLY **Date:** May 5, 2026 **Analyst:** Olayinka Oyetade **Severity:** HIGH **Status:** RESOLVED

1. EXECUTIVE SUMMARY

A simulated persistence attack was executed against a controlled Windows 10 endpoint within a home lab environment. The threat actor (simulated) planted a malicious registry Run key entry designed to execute a payload (`malware.exe`) on every user login. The attack was successfully detected by Sysmon (Event ID 13) and confirmed via Splunk within approximately 5 minutes of execution. An automated email alert was triggered, confirming the end-to-end detection pipeline is functional.

2. INCIDENT TIMELINE

Time (Relative)	Event
T+00:00	Attacker executes PowerShell command on compromised Windows 10 VM
T+00:01	Sysmon captures registry modification — Event ID 13 logged
T+00:02	Splunk ingests log from Windows VM into <code>endpoint</code> index
T+00:04	Saved search detection rule matches suspicious pattern
T+00:05	Automated email alert dispatched to analyst inbox
T+00:05	Analyst receives notification and begins investigation
T+00:07	Incident confirmed, contained, and documented

3. ATTACK DESCRIPTION

3.1 Attack Vector

The attacker gained code execution on a Windows 10 endpoint (simulated) and used PowerShell to write a malicious entry into the Windows Registry Run key — a location Windows reads automatically at every user login to launch programs.

3.2 Command Executed

```
New-ItemProperty -Path "HKCU:\Software\Microsoft\Windows\CurrentVersion\Run" `
    -Name "WindowsUpdate" `
    -Value "C:\Users\Public\malware.exe" `
    -PropertyType String `
    -Force
```

3.3 Attacker Objective

By writing to the Run key, the attacker ensures that `malware.exe` is launched automatically every time the user logs into Windows — even if the endpoint is rebooted or antivirus removes the payload from disk, the registry entry remains until manually cleaned, causing re-infection on next login.

3.4 Why This Technique Is Dangerous

- Survives reboots automatically
- Disguised under a legitimate-sounding name ("WindowsUpdate")
- Does not require administrator privileges (HKCU key)
- Commonly used by: Emotet, TrickBot, Remcos RAT, and multiple ransomware families

4. DETECTION

4.1 Detection Source

Field	Value
Detection Tool	Splunk Enterprise
Data Source	Sysmon for Windows

Event ID	13 (Registry Value Set)
Index	endpoint
Sourcetype	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

4.2 Key Log Fields Captured

Field	Value
EventCode	13
RuleName	technique_id=T1547.001, technique_name=Boot or Logon Autostart Execution
Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
TargetObject	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\WindowsUpdate
Details	C:\Users\Public\malware.exe
User	WINDOWSVM10\mrdaniel198

4.3 Detection Query (SPL)

```
index=endpoint EventCode=13 "powershell.exe" "CurrentVersion\\Run"
| table _time, host, User, Image, TargetObject, Details
| sort -_time
```

4.4 Alert Configuration

- **Alert Name:** Suspicious Registry Run Key Modification
- **Trigger Condition:** Any result returned by detection query
- **Alert Action:** Email notification to analyst
- **Time to Alert:** < 5 minutes from attack execution

5. MITRE ATT&CK MAPPING

Field	Value
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Tactic	Persistence (TA0003)
Technique	T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder
Sub-Technique	Registry Run Keys
Platform	Windows
Permissions Required	User
Data Sources	Windows Registry, Process Creation, Command-Line Parameters

6. IMPACT ASSESSMENT

Category	Assessment
Confidentiality	HIGH — Persistent malware could exfiltrate sensitive data
Integrity	HIGH — Malware could modify or destroy system files
Availability	MEDIUM — Could disrupt services or encrypt data (ransomware scenario)
Business Impact	HIGH — Undetected persistence leads to long-term compromise

7. CONTAINMENT & REMEDIATION

Immediate Actions Taken

1. Identified the malicious registry key via Splunk investigation
2. Removed the registry entry:

```
Remove-ItemProperty -Path "HKCU:\Software\Microsoft\Windows\CurrentVersion\Run" -
```

3. Confirmed removal by re-querying Splunk for subsequent Event ID 13 activity

Recommended Production Actions

- Isolate the affected endpoint immediately upon alert

- Preserve disk image for forensic analysis
 - Scan for additional persistence mechanisms (scheduled tasks, services, startup folders)
 - Reset credentials for affected user account
 - Notify incident response team and escalate per playbook
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8. EVIDENCE

Evidence Item	Description
Sysmon Event ID 13 log	Raw log captured in Splunk showing registry modification
Splunk detection screenshot	Dashboard showing query results and alert
Email notification screenshot	Proof of automated alerting pipeline
PowerShell command screenshot	Evidence of simulated attack command

9. ANALYST NOTES

This simulation confirmed that behavioral detection (catching the action of writing to a Run key) is more reliable than signature-based detection (trying to identify the specific malware file). The detection fired regardless of the filename or payload — it detected the attacker behavior, not the specific tool, which makes it significantly harder to evade.

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